



Digital Transformation for Micropropagation

Lab to Greenhouse: Intelligent Plant Tissue Culture Management System

Why we need a system

Micropropagation at scale fails when tracking stays in notebooks. Thousands of vessels, invisible lineages, and late discovery of errors.



Lineages lost in notebooks

1000s of vessels with no digital genealogy from Stage I to IV



Contamination root-cause unknown

Bacterial / fungal events untracked to media, hood, or operator



Media prep errors & chemical waste

Manual math, no expiry logging, no auto-deduction



No yield forecasting

Orders delayed, split ratios not tracked, no multiplier history

2,500+

Active jars / week
per lab bench

18%

Avg contamination
without root-cause data

4 Stages

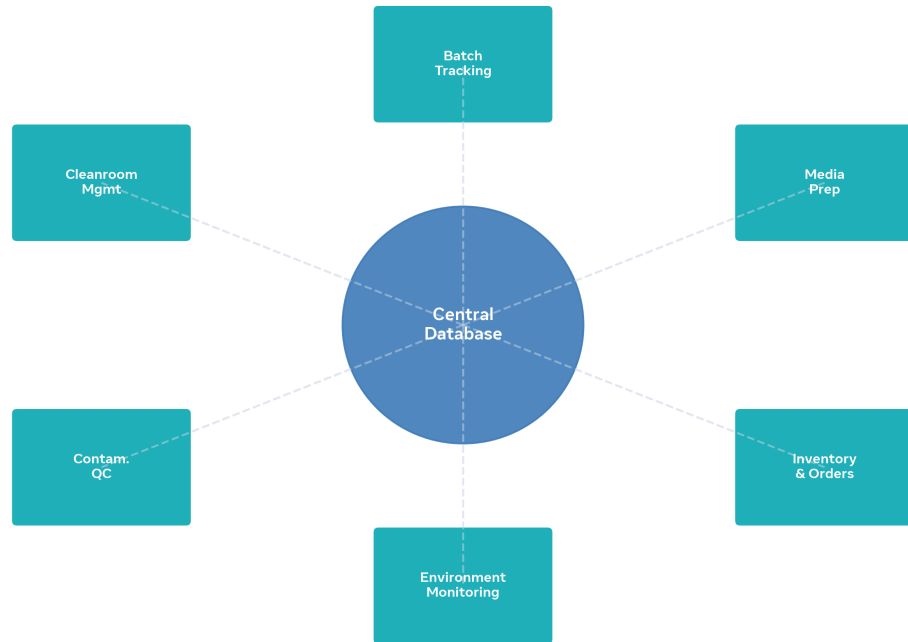
Initiation → Multi →
Rooting → Acclim.

0

Traceability today
from mother → dispatch

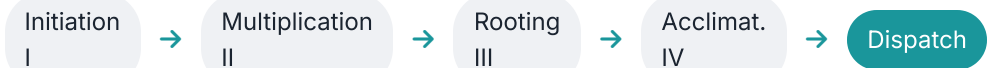


Solution: End-to-end digital traceability from Stage I to IV
— every vessel, media bottle, and transfer QR-scanned.



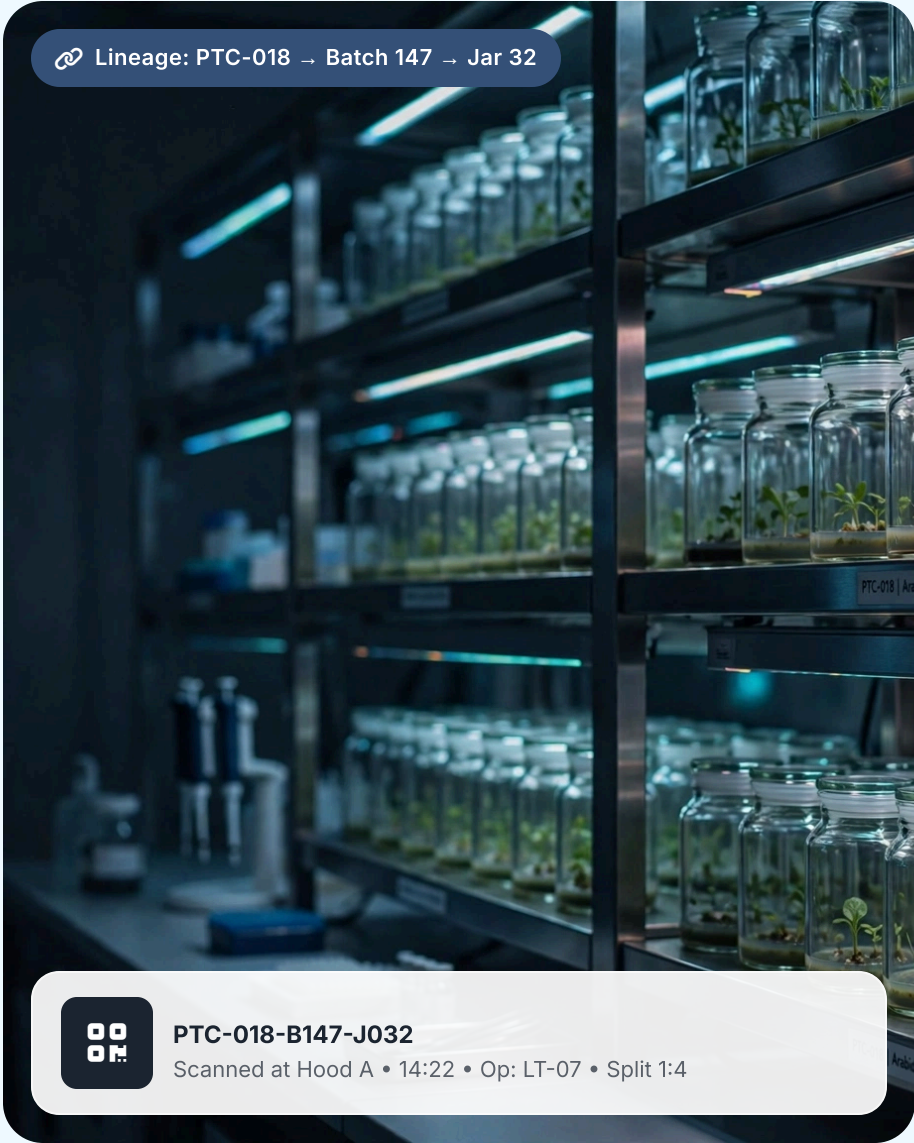
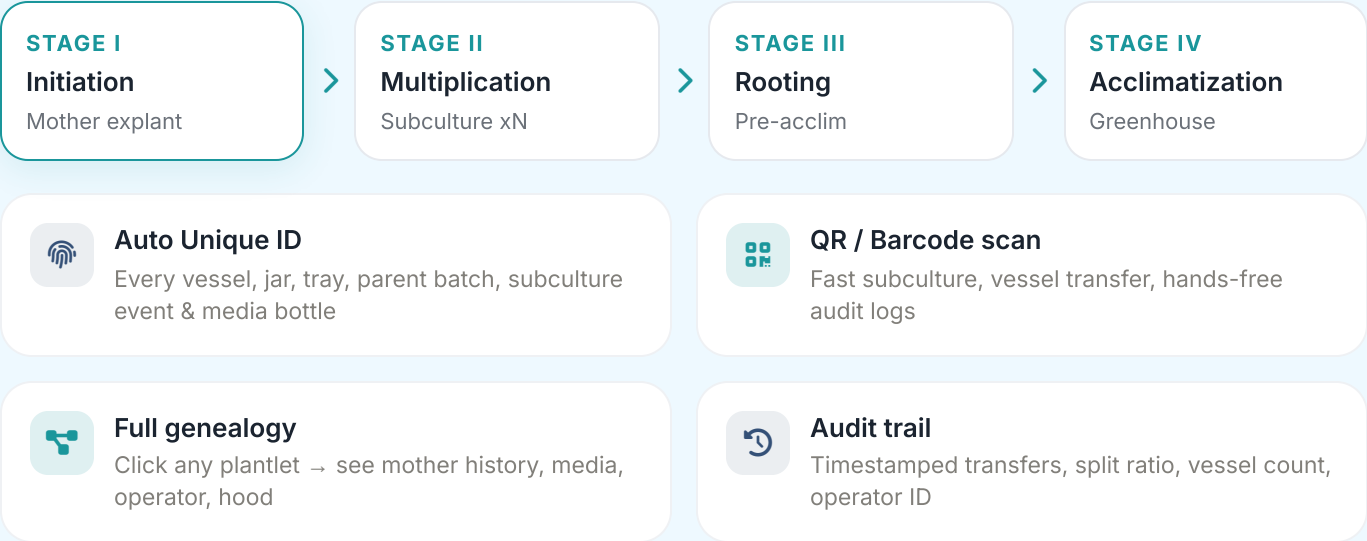
Architecture at a glance

- 1 Batch Tracking**
QR / Barcode lineage
- 2 Media Prep**
Recipe + autoclave
- 3 Cleanroom Mgmt**
Laminar hood logs
- 4 Contamination QC**
Root-cause analytics
- 5 Environment**
Temp / RH / PAR / IoT
- 6 Inventory & Orders**
Stock to dispatch



Central database connects every scan, media batch, and environment reading — one source of truth from mother plant to customer.

Module 1: Batch & Container Tracking



Media preparation & recipe management

No manual math errors. Every bottle traced to autoclave, operator, and expiry.

 Recipe Builder — standardized

MS / WPM
Basal Media

Stock Sol.
Macro / Micro

PGRs
BAP, NAA, Kin

Vitamins
B1, Myo-inositol

Gelling
Agar, Phytigel

pH 5.8
Auto-checked



Vol_{stock} = Target Vol × Desired Conc / Stock Conc
Example: 2.5 mL BAP stock (1 mg/mL) for 1L @ 2.5 mg/L — calculated automatically

 Autoclave & inventory linked

-  **Autoclave batch logging**
Date, Temp 121°C, Pressure 15 psi, Expiry 14 days, Operator ID
-  **Media bottle QR generated**
Links recipe version → autoclave run → vessel batch
-  **Inventory deduction on creation**
Raw chemicals auto-deducted with lot tracking & low-stock alert



Traceable: Click any vessel → see exact media batch, recipe, and who autoclaved it.

Every hood scan is logged — no gloves off, no data lost

MODULE 3 • CLEANROOM MGMT

 Batch IN

 Split

 Batch OUT



Laminar flow hood workstation logging

Operator ID, vessel count, split ratio, timestamp — captured via QR/barcode scan for fast transfer and audit trail.



Hands-free scanning workflow

Designed for gloved operators: mobile + handheld scanners, auto timestamps, batch IN/OUT linked to hood ID.



Performance metrics per operator & workstation

Track throughput, contamination correlation, and training needs — not just batches.

Multiplier calculator projects expected yield

HISTORICAL RATE

3.8x

VESSELS IN

48 jars

SPLIT RATIO

1:4

PREDICTED OUT

182

Yield forecast

+12% vs last subculture

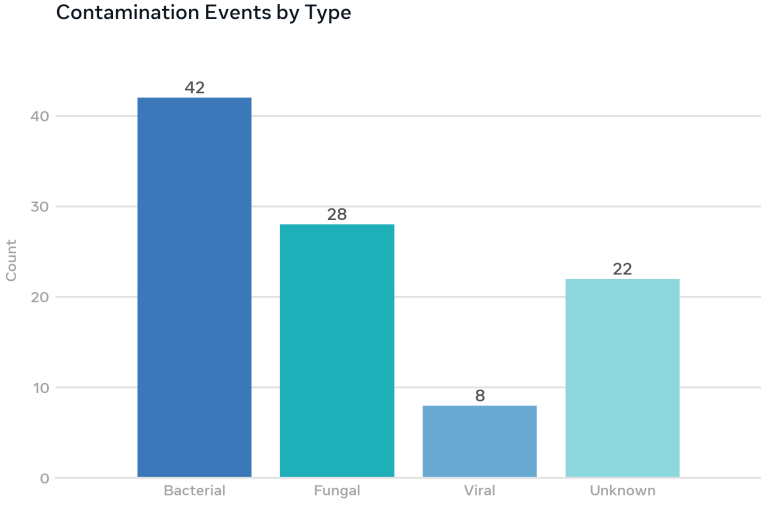


Uses last 3 subculture cycles per lineage to forecast Stage II ... III needs for order fulfillment.

Module 4: Contamination & Quality Control

 LIVE CONTAMINATION LOG

By Type • Last 30 days



 Vessel + Media Batch

 Operator + Hood ID

 Room / Location



Analytics dashboard pinpoints root cause

Filter contamination by media batch failure vs operator error vs laminar hood malfunction. Heatmap by room, batch, and user — see if 42% bacterial spike traces to one autoclave run.



Early warning system for spikes

Auto-alert when daily rate exceeds 3-sigma of historical baseline. Triggers audit of hood filters, operator retraining, or media hold.



Discard & mortality tracking

One-scan waste logging: reason, count, cost per vessel. Mortality rate by stage — links to growth-room alerts and yield forecast.



18% → 4%

Target reduction with root-cause closure



QR scan

Log event in 3 sec with photo proof

Module 5: Environmental & Growth Room Monitoring



Temperature
22–26°C logged



Humidity
RH % every 5m



PAR / Light
 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$



Photoperiod
16h / 8h cycle



Manual or automatic. Tablet entry in room, or IoT sensors via MQTT / REST APIs — unified timeline.



Alerts for out-of-range. Slack/email/panel alert when thresholds breach; tied to batch location.



Correlate growth vs environment. Link multiplication rate to GR logs to refine protocols.



Every reading timestamped to Stage I-IV location — traceable in batch genealogy view.

● **Out-of-range protection:** Temp 24.2°C $\pm$ 1°C | RH 68% | PAR 65 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ | Photoperiod 16/8. Auto-alert if drift.

Module 6: Inventory & Order Management

From chemical lot to customer dispatch — stock, expiry, and ready-to-ship in one queue, linked to Stage IV acclimatization.



5 LOW-STOCK

Raw materials stock control

Chemicals, culture vessels, tools, PPE · lot-tracked with low-stock alerts & reorder suggestion based on media plan.

 MS salts — 2.1 kg · reorder at 1 kg

Glass jars 250ml — 84 left

127
SKUs tracked

₹1.2L
Cost per 1000 plantlets



EXPIRY SYNC

Media & consumable expiry tracking

Autoclaved media, PGR stocks, vitamins — expiry auto-calculated, blocked from use after date.

- ✓ Media batch MB-241 blocking after 14 days
- ✓ BAP stock 1mg/mL · exp 2026-04-12



STAGE IV → ORDER

Order fulfillment linked

Ready acclimatized plants auto-linked to customer sales orders — no spreadsheet handoff.

 Tray TR-IV-889 → Order #SO-1047 (Banana G9)

 Customer history & survival rate tracked



12 READY TO SHIP

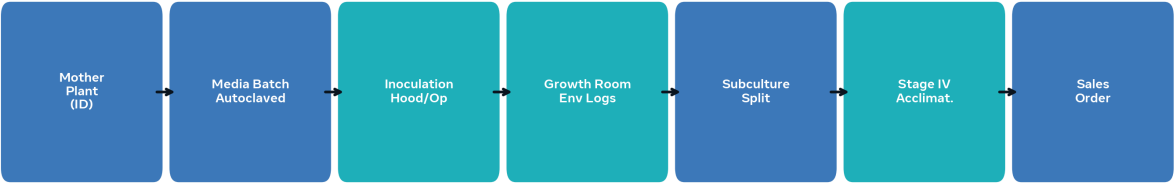
Dispatch & ready-to-ship queue

Dispatch tracking, packing list, courier ID — scanned at greenhouse exit for audit.

48h
Avg Stage IV to dispatch

98.2%
On-time fulfillment

END-TO-END TRACEABILITY — 7 LINKED CHECKPOINTS



Mother ID

Media Batch

Inoculation

Growth Room

Subculture

Stage IV

Sales Order

Full lifecycle: One scan tells everything

Every transfer is QR-linked across 6 modules — no gap from mother plant to dispatch.



QR-scanned at each handoff

Mother → Recipe + Autoclave → Hood/Operator → Growth Room env logs → Split → Stage IV → Order



Timestamped & auditable

Who, where, when: vessel count, split ratio, media expiry, hood ID, operator — immutable log



Click any plantlet → full genealogy

Trace back to mother explant, media batch, contamination checks, and environment history in one view



Compliance-ready: complete audit trail for QC, customer COA, and yield forecasting — zero notebook dependency.

From data to decisions



Multiplication rate forecasting

Predict Stage II yield using last 3 subcultures per lineage and split ratio history.



Contamination heatmap by room / operator / media

Spot if 42% bacterial events trace to Hood B, Operator LT-07, or Media Batch M-114.



Yield projection vs actual orders

Link Stage IV acclimatized queue to sales orders — flag shortfall 14 days early.



Operator efficiency & training needs

Throughput and contamination rate per workstation drives targeted retraining.



Inventory cost per plantlet

Media + labor + loss cost rolled up to exact margin per cultivar.

Intelligence dashboard – live

● updated 2m ago

AVG MULTI. RATE
3.8x
+0.4 vs last month

CONTAMINATION
4.2%
↓ 11% alert threshold

YIELD ACCURACY
94%
forecast vs actual

COST / PLANTLET
\$0.42
\$0.06 saved



Contamination by source

H-A 2%	H-B 8%	H-C 1%	H-D 3%
OP-03 1%	OP-07 6%	M-114 9%	M-115 2%
GR-01 2%	GR-02 5%	Lot-C 1%	Lot-D 4%

Darker = higher. Click cell to see linked vessels and media bottles.

Lab Manager:
throughput & schedule

QC Manager:
root-cause closure

Operations:
orders & margin

⚡ BUILT FOR THE REAL LAB

Tech stack & integration ready



Barcode / QR anywhere

Mobile + handheld scanners, hands-free logs, offline capture in cleanroom



IoT ready: MQTT & REST

Temp / RH / PAR sensors push to growth room timeline automatically



Role-based access

Lab Tech, QC, Manager, Inventory, Sales — scoped actions & dashboards



Audit trail + cloud backup

Compliance-ready logs, cloud sync, offline mode for laminar hoods

NEXT STEPS

● Prototype scanning flow



🧪 Recipe builder MVP



👤 Pilot in Growth Room 1

Q2 rollout